

Question: 1

```

SELECT e. Employee ID,
       e. FirstName,
       e. LastName,
       e. Department,
       e. Salary,
       p. Project ID,
       p. Project Name,
       p. Budget,
       p. Status

```

```

FROM Employee e

```

```

INNER JOIN Projects p

```

```

ON e.EmployeeID = p.EmployeeID;

```

Employee ID	FirstName	LastName	Department	Salary	Project Name	Project ID	Budget	Status
1	John	Doe	IT	70 000	AI Development	101	100 000	Completed
2	Alice	Smith	HR	60 000	Employee training	102	50 000	Ongoing
3	John	Doe	IT	70 000	Cyber Sec Audit	103	75 000	Pending
4	Bob	Johnson	Finance	75 000	Financial Analysis	104	90 000	Ongoing
5	Emma	Wilson	Sales	65 000	Market Expansion	105	65 000	Completed
6	Michael	Clark	Finance	80 000	Risk Management	106	80 000	Pending

Question: 2

```

SELECT e. Employee ID,
       e. First Name,
       e. Last Name,
       e. Department,
       e. Salary,
       p. Project ID,
       p. Project Name,
       p. Budget,
       p. Status

```

```

FROM Employee e

```

```

LEFT JOIN Projects p

```

```

ON e.EmployeeID = p.EmployeeID;

```

Employee ID	FirstName	LastName	Department	Salary	Project ID	Project Name	Budget	Status
1	John	Doe	IT	70 000	101	AI Development	100 000	Completed
1	John	Doe	IT	70 000	103	CyberSec Audit	75 000	Pending
2	Alice	Smith	HR	60 000	102	Employee training	50 000	Ongoing
3	Bob	Johnson	Finance	75 000	104	Fin Analysis	90 000	Ongoing
4	David	Brown	IT	72 000	NULL	NULL	NULL	NULL
5	Emma	Wilson	Sales	65 000	105	Market Exp	65 000	Completed
6	Michael	Clark	Finance	80 000	106	Risk Man	80 000	Pending

Question : 3

```

SELECT p. ProjectID,
       p. ProjectName,
       p. Budget,
       p. Status,
       e. EmployeeID,
       e. FirstName,
       e. LastName,
       e. Department,
       e. Salary,
FROM Employee e
RIGHT JOIN Projects p
ON e.EmployeeID = p.EmployeeID;

```

ProjectID	ProjectName	Budget	Status	EmployeeID	FirstName	LastName	Department	Salary
101	AI Development	100000	Completed	1	John	Doe	IT	70000
102	Employee training	50000	Ongoing	2	Alice	Smith	HR	60000
103	CyberSec Audit	75000	Pending	1	John	Doe	IT	70000
104	Fin Analysis	90000	Ongoing	3	Bob	Johnson	Finance	75000
105	Market Expansion	65000	Completed	5	Emmy	Wilson	Sales	65000
106	Risk Management	80000	Pending	6	Michael	Clark	Finance	80000

Question : 4

```

SELECT e. EmployeeID,
       e. FirstName,
       e. LastName,
       e. Department,
       e. Salary,
       p. ProjectID,
       p. ProjectName,
       p. Budget,
       p. Status
FROM Employee e
FULL OUTER JOIN Projects p
ON e.EmployeeID = p.EmployeeID;

```

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	70000	101	AI Development	100000	Completed
1	John	Doe	IT	70000	103	CyberSec Audit	75000	Pending
2	Alice	Smith	HR	60000	102	Employee Training	50000	Ongoing
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing
4	David	Brown	IT	72000	NULL	NULL	NULL	NULL
5	Emmy	Wilson	Sales	65000	105	Market Expansion	65000	Completed
6	Michael	Clark	Finance	80000	106	Risk Management	80000	Pending

Question : 5

```
SELECT City AS Location
FROM Employees

UNION
```

```
SELECT Status AS Location
FROM Projects;
```

Location
New York
Los Angeles
Toronto
London
Sydney
Completed
Ongoing
Pending

Question 6

```
SELECT City AS Location
FROM Employees
```

UNION ALL

```
SELECT Status AS Location
FROM Projects;
```

Location
New York
Los Angeles
Toronto
London
Sydney
New York
Completed
Ongoing
Pending
Ongoing
Completed
Pending

Question : 7

```
SELECT EmployeeID,
       FirstName,
       LastName,
       Department,
       Salary
FROM Employees
WHERE Salary > 70000;
```

EmployeeID	FirstName	LastName	Department	Salary
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Michael	Clark	Finance	80000

Question : 8

```
SELECT EmployeeID,  
        First Name,  
        LastName,  
        Department,  
        Salary  
FROM Employees  
WHERE Department IN ('IT', 'Finance');
```

Employee ID	First Name	Last Name	Department	Salary
1	John	Dee	IT	70 000
3	Bob	Johnson	Finance	75 000
4	David	Brown	IT	72 000
6	Michael	Clark	Finance	80 000

Question : 9

```
SELECT ProjectID,  
        Project Name,  
        Budget,  
        Status  
FROM Projects  
WHERE Status <> 'Completed';
```

Project ID	Project Name	Budget	Status
102	Employee training	50 000	Ongoing
103	CyberSec Audit	75 000	Pending
104	Financial Analysis	90 000	Ongoing
106	Risk Management	80 000	Pending

Question : 10

```
SELECT Project ID,  
        Project Name,  
        Budget,  
        Status  
FROM Projects  
WHERE Budget > 70 000  
AND Status <> 'Completed';
```

Project ID	Project Name	Budget	Status
103	CyberSecurity Audit	70 000	Pending
104	Financial Analysis	90 000	Ongoing
106	Risk Management	80 000	Pending

Question : 11

```
SELECT EmployeeID,  
       First Name,  
       LastName,  
       Department,  
       Salary,  
       City  
FROM Employees  
WHERE City IN ('New York', 'Toronto')  
ORDER BY Salary DESC;
```

EmployeeID	FirstName	LastName	Department	Salary	City
6	Michael	Clark	Finance	80 000	New York
3	Bob	Johnson	Finance	75 000	Toront
1	John	Doe	IT	70 000	New York

Question : 12

```
SELECT EmployeeID,  
       FirstName,  
       LastName,  
       Department,  
       Salary  
FROM Employees  
ORDER BY Salary DESC  
LIMIT 3;
```

EmployeeID	FirstName	LastName	Department	Salary
6	Michael	Clark	Finance	80 000
3	Bob	Johnson	Finance	75 000
4	David	Brown	IT	72 000

Question : 13

```
SELECT Department  
       sum(Salary) AS TotalSalary  
FROM Employees  
GROUP BY Department  
ORDER BY TotalSalary
```

Department	Total Salary
Finance	155 000
IT	142 000
HR	60 000
Sales	65 000

Question : 14

```
SELECT City,  
       AVG(Salary) AS AverageSalary  
FROM Employees  
GROUP BY City  
HAVING AVG(Salary) > 65000;
```

City	Average Salary
New York	75000
Toronto	75000
London	72000

Question : 15

```
SELECT Department,  
       COUNT(EmployeeID) AS EmployeeCount  
FROM Employees  
GROUP BY Department  
HAVING COUNT(EmployeeID) > 1;
```

Department	Employee Count
IT	2
Finance	2

Question : 16

```
SELECT Status,  
       COUNT(ProjectID) AS ProjectCount  
FROM Projects  
GROUP BY Status  
HAVING COUNT(ProjectID) >= 2;
```

Status	Project Count
Completed	2
Ongoing	2
Pending	2

Question : 17

```
SELECT e.EmployeeID,  
       e.FirstName,  
       e.LastName,  
       SUM(p.Budget) AS TotalProjectBudget  
FROM Employees e  
INNER JOIN Projects p  
ON e.EmployeeID = p.EmployeeID  
GROUP BY e.EmployeeID, e.FirstName, e.LastName  
HAVING SUM(p.Budget) > 150000;
```

EmployeeID	FirstName	LastName	TotalProjectBudget
1	John	Doe	175000